

Essence of Internet of Medical Things in Geriatric Healthcare

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Abstract—The healthcare is a notable discipline towards continuous inventions and sustainable developments worldwide. Because of swift technological advancement in industry to imagine the healthcare without 'Artificial Intelligence', 'Cloud Computing', 'Internet of Things', will be a miss in better and efficient prognosis or diagnosis on early stages. The healthcare industry benefited from aforementioned technologies from two decades but there are areas which has yet to avail services i.e. Geriatric Healthcare. The paper essentially focused on geriatric healthcare services, required technological amendments, challenges and possible solutions with Internet of Things for the betterment of health services.

Index Term: Artificial Intelligence (AI), Assisted Living, Healthcare, Internet of Things (IoT), Software Development Kit (SDK), Sensor, Smart Devices, Telemedicine, Wearable.

I. INTRODUCTION

The rapid progress in clinical science and technology has had a significant impact on increasing the average life expectancy of individuals worldwide. Consequently, there has been a notable surge in the geriatric population. In 2015, the number of elderly individuals accounted for approximately 8.5% of the global population, with projections indicating a rise to 12% and 16.7% by 2030 and 2050, respectively. Elderly individuals are more susceptible to various health conditions such as diabetes, hypertension, asthma, and chronic illnesses compared to other age groups. Therefore, this demographic requires heightened attention concerning medication, treatment, and care, particularly if they opt for an independent lifestyle. The primary challenge in accessing quality healthcare services lies in the escalating costs. Additionally, the elderly population may face difficulties in physically visiting healthcare facilities each time they encounter a health issue. The cause could be attributed to either the rise in expenses or the lack of access to a well equipped healthcare facility. This situation has motivated different research groups to explore alternative options that

can lower costs without compromising the quality of healthcare services provided to patients. Thanks to the widespread utilization of advanced technologies, internet services, and sensors, individuals can now access a variety of healthcare services from the comfort of their own homes. This enables elderly individuals to maintain their independence while still receiving high-quality clinical care in their own residences. However smart healthcare technology now provisioned to majority but few domain still require more support such as geriatric healthcare which missing due to many factors viz. cost, reliability, acceptance etc. The said domain has prompted various research groups to investigate alternative solutions that can reduce costs without compromising the quality of healthcare services offered to patients. The paper is devoted to provide detailed overview on geriatric healthcare with smart technology. The rest of the paper is segregated into few sections. The section II briefed about requirement of health monitoring, section III interpreted about impact of smart technology in healthcare, section IV precisely tells about existing improvement in technology for geriatric care and section V is dedicated to provide an overview on problems, future research dimensions for producing tangible solutions.

II. HEALTH MONITORING IN GERIATRIC POPULATION

Aging is a natural biological process that causes diseases associated with aging to arise. These illnesses are brought on by because of external or endogenous influences in morphological and functional alterations in all organs and systems [1]. The Physiological alterations could lead to the onset of long-term conditions including depression and osteoporosis. These illnesses are frequently concealed and persist unusual for a longer period of time. When it comes to the elderly population, the illnesses spread more quickly and could become difficult to understand. Polypharmacy is a side effect of treating diseases connected to aging in contrast to the middle-aged.

A. Sensory impairments

The quality of life of elderly is greatly impacted by both visual and hearing impairments. Depression, social isolation, trouble communicating, and limited mobility can all result from hearing loss. On the other side, despair, physical infirmity, and increased social isolation are linked to visual impairment. Managing everyday tasks and drug regimes is much more difficult for elderly people who have both visual and hearing impairments [2] [3] [4].

B. Falls and their after-effects

A large percentage of the older population suffers from falls on a yearly basis, making it a widespread problem. Elderly individuals, particularly those experiencing difficulties with their stride, are more vulnerable to falls, which can lead to severe injuries such as fractures and dislocations of joints. Falls frequently result in chronic pain, diminished function, and a loss of independence, making long-term care and therapy necessary. Due to greater death rates among men, women make up the bulk of the senior population and are especially vulnerable to falls [5] [6] [7].

III. IMPACT OF TECHNOLOGY ON GERIATRIC HEALTH MONITORING

Artificial intelligence, Cloud computing, and IoT all together are significant technologies having essential roles in healthcare in the creation of an effective system that can improve the quality of life for the elderly population. These technologies are employed in the construction of portable gadgets like cell-phones, rehabilitative equipments, and wearable devices towards support in public health services in isolated areas. When these technologies are applied effectively, it will lessen the medical burden on physicians and health center.

A. Remote monitoring with wearable

Sensors provide real-time monitoring capabilities which support in cardiac health monitoring, fall detection, sleep pattern tracking, and activity monitoring in the elderly [8] [9] [10]. Better health outcomes are facilitated by these technologies, which allow for prompt alarms to be sent to caregivers. Innovative wearable technology and commercially available gadgets like headbands and smart-watches have the potential to improve senior care [11] [12] [13].

B. Ambient assisted living systems

Ambient assisted living (AAL) systems support independent living while monitoring health parameters and identifying irregularities. They do this by integrating information and communication technology into the everyday life of the aged. These technologies provide communication, activity monitoring, and environmental correlation for better care and lower healthcare expenses [14] [15] [16] e.g. robotic-assisted services and the 'Keep In Touch (KIT)' technology.

C. Telemedicine and mobile healthcare services

Telemedicine allow medical practitioners to communicate with senior patients with audio-visual means, which makes remote diagnosis, treatment, and consulting easier. Mobile healthcare services improve accessibility and efficiency in the delivery of care by preserving medical data through cell-phones and cloud-based storage. These technologies allow for ongoing monitoring and communication with healthcare practitioners, empowering senior citizens to take charge of their own health [18] [19] [20] [21]. This process improves the availability of clinical data and the proficiency in proposed or designed frameworks or applications. It has been accounted that more seasoned grown-ups generally favour a cell phone over other electronic gadgets like PCs, versatility of these gadgets and the simplicity of availability might be the purpose for their colossal prevalence. In notable IoT oriented frameworks or applications, the cell phone is utilized as an entryway for information correspondence among emergency clinics, drug stores, clinical specialists, and patients.

IV. MECHANICAL INNOVATION AND IoT APPLICATIONS IN GERIATRIC CARE

Mechanical innovation has shifted old medical services by understanding how likely the utilization of human robotization, human computer interaction will helps the geriatric population in day to day movement, cautions the patients require about their medical problems, guarantees the security of the patient, and offers social help. There are three unique kinds of robots that could be utilized in geriatric consideration which incorporate family robots, friend robots, and assistive robots [22]. The combination of robots in IoT based medical care helps older handicapped people to perform actual assignments with mechanical innovation (see Figure 1). It is feasible to make a feeling of the actual presence of the parental figures and specialists from a distance that can essentially interface with the patients through a mechanical body [23] [24], likewise it announced the proficient utilization of telepresence automated innovation in monitoring ECG, video conferencing, and reminding patients to take medication [23] etc.

In [25] IoT, remote correspondence were coordinated into a wheelchair for ongoing checking or visualize the general climate with the assistance of an exhibit of camera which is assisting with finding more secure route undertakings. In a new report, specialists have combined advanced mechanics with other detecting advances in an IoT based AAL climate to give a higher human-robot communication. Such frameworks can figure out the patient's necessities furthermore and can act as needs be in a more versatile way [17]. IoT has changed the healthcare with cutting edge detecting advances, correspondence conventions, and information investigation procedures. This has affected the existence of geriatric individuals by working on their personal satisfaction and giving a extensive variety of medical care applications. This prominent technology has brought about revolutionary

who were followed by Microsoft Kinect SDK. The highlights of the paired picture were separated utilizing the histogram of situated slope (Hoard) technique. The status of the fall was assessed utilizing the help vector machine (SVM) calculation. Mix of IoT for action/fall recognition achieves many benefits and has been broadly utilized as far as remote access. The utilization of wearable sensors in these proposed frameworks is can be made sense, by the way that wearing a detecting gadget all is unimaginable. Besides, the utilization of these wearable sensors moreover is imperfect is incorporates the deficiency of wearable gadgets, upkeep trouble, uneasiness in wearing, and lower battery duration [38]. Consequently, gadget free frameworks should be created in the future with negligible utilization of wearable sensors.

C. Persistent illnesses observing

Persistent illnesses are one of the main sources of death in old individuals [39]. According to a clinical study [40], around 80% of geriatric individuals matured over 65 experience the ill effects of somewhere around one persistent sickness. concerning to diabetes, cardiovascular illnesses, disease, discouragement, alzheimer's illness, osteoporosis, lung sickness, kidney illness, Parkinson's sickness, dementia etc. It is challenging for old individuals to deal with themselves in presence of the previously mentioned clinical circumstances. Further, the determination and observing of persistent illnesses need non stop exertion. In such manner, different IoT driven gadgets have shown expected in managing with these infections. A few examinations [41] [42] [43] [44] [45] in the past have proposed IoT based answers for work on the expectation for everyday comforts of geriatric patients with the previously mentioned infections. The study stated that remote observing in geriatric care can diminish the death rate alongside hospitalization rate and conveyed quality ongoing illness treatment [45]. In [44] the creators proposed a framework to screen essential signs that assistance to identify different ongoing infections of geriatric patients utilizing a scope of wearable sensors and information mining approaches were taken on for preparing the such framework.

V. PRESENT PROBLEMS, DIFFICULTIES, AND FUTURE DIMENSIONS

The creation of a healthcare system that will be effective for the elderly population needs to be predicated on three significant aspects - effective use, quick response with efficient communication, and an intuitive user interface with less cost. The available healthcare devices are unable to offer reliable connectivity and quick access, therefore, the design and development of systems with leading technology which will address such problems is a primary requirement in healthcare. The IoT has revolutionized healthcare services by offering reliable solutions in a range of healthcare issues. In spite of its broader uses, this scientific field still faces numerous difficulties i.e. the majority of senior-care options now on the market have disregarded the functional constraints

brought on by aging and declining cognitive functionality. The categories of challenges or issues in old age are person-specific and require the creation of adaptive systems as per the requirements. In addition, older individuals have trouble using modern devices and require more time to become application friendly, so to taking into account the extensive knowledge of behavioural analysis as a feedback mechanism during the creation of the IoT based healthcare system is one possible way to address the aforementioned problem. Standardization in the aforementioned disciplines or proposed systems is also essential i.e. fostering standards on design/developed medical technology could offer essential security and enable strong software development security solutions. Despite the fact that the complexity of the standardization process is increasing due to the advancements in IoT healthcare applications, no specific study that addressed these needs and challenges was found in the literature. Moreover an increased amount of medical information is produced by the IoT applications which links numerous medical devices and sensors, the produced data are dispersed over many repositories and the requirement of securing this information is a challenges task. Since patient sensitive and private information is contained in healthcare repositories so careful handling of such information is primary requirement. In order to fulfil former reliable and robust security application should be designed, which is missing in healthcare. Several physicians and caregivers involved are from various geographic regions in personalized healthcare services which necessitate effective communication between these service providers. For this reason, the future of AI data driven model that can both detect and integrate several data sources and offer an effective data-handling mechanism would be necessary for the healthcare system. Additionally, the proposed healthcare system may use patients' authorization before utilizing their medical records which is somehow missing in majority of existing solutions. Below are the major challenges which requires attention and possible addressing:

- Intuitive, reliable and cost-effective healthcare applications for geriatric population
- Considerable and less-computationally intensive security solutions in healthcare applications
- Standardization on healthcare data and applications to support interoperability aspect on information sharing and its uses
- Patient control on own healthcare information with require data-sharing policies

VI. CONCLUSION

The industry professional started to investigate possible technological options to address the issues of geriatric healthcare services with the uses of potential Internet of Things and Wearable technology, which remains intact from years. Even though the advent technology have potential to provide the solutions but worldly acceptable, reliable and efficient strategies are still far from achieve in geriatric healthcare. The paper briefed about recent developments

in healthcare along with areas which are yet to be touched which can be benefited with IoT due to its potential towards providing tangible solution.

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